

Elyes Darouich

AI Engineer — Agents, RAG & LLM Systems

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github.com/ElyesD1 | github.com/ElyesDarouich | linkedin.com/in/elyes-darouich | Portfolio | Misalignment write-up

PROFESSIONAL SUMMARY

AI engineer (National Engineering Degree, ESPRIT 2026) building production agentic systems, guardrailed RAG pipelines, and private on-premise LLM deployments. Delivered an AI ISO-compliance auditor at Talan — clause-by-clause verdicts with cited evidence across 110 clauses on Llama 3.3 70B — retained as an internal demonstrator. Builds and evaluates local LLM systems end to end: a 1,596-test suite, a pre-registered misalignment evaluation, LoRA fine-tuning on Apple MLX.

TECHNICAL SKILLS

AI / ML: Multi-Agent Orchestration, LangGraph, RAG, LLM Guardrails & Evaluation, LoRA Fine-Tuning (Apple MLX), On-Premise LLM Deployment (Ollama), MCP, Claude Agent SDK, Prompt Engineering, Vector Search (ChromaDB, Qdrant, sqlite-vec), Embeddings

Languages: Python, TypeScript, JavaScript, Swift, Java, C++, Solidity, SQL

Backend & Data: FastAPI, NestJS, Flask, Node.js, Celery, REST, WebSockets, PostgreSQL, SQLite, MongoDB, Redis

Frontend & Mobile: React, Next.js, SwiftUI, Flutter, Tailwind CSS

DevOps & Web3: Docker, GitHub Actions CI/CD, Linux, Vercel, Zero-Knowledge (Groth16, Circom, snarkjs), Hyperledger Besu

EXPERIENCE

End-of-Studies AI Engineer (PFE) — Talan Tunisia

Feb 2026 – Jun 2026

Tunis, Tunisia

- Architected “SMI Virtual Auditor,” an AI ISO-compliance audit platform producing clause-by-clause verdicts with cited evidence (ISO 9001/14001/27001, 110 clauses) via a 3-pass RAG pipeline on Llama 3.3 70B (Groq, GitHub Models, NVIDIA NIM).
- Engineered a 7-layer guardrail system (groundedness, fidelity, contradiction, confidence capping) that suppresses hallucinations, grounding verdicts via local bge-m3 embeddings on Ollama + ChromaDB.
- Built a clean-architecture FastAPI backend (JWT RS256, RBAC, Celery, PostgreSQL/Redis/MinIO, Docker, 108 tests) and a React 18 + TypeScript SPA with streaming RAG chat (SSE) and PDF export.

Software Engineering Intern — Talan Tunisia

Jul 2025 – Sep 2025

Tunis, Tunisia

- Built “ProjectFlow” (Flutter + NestJS): sub-150ms APIs for 100+ concurrent users at 78% test coverage, real-time Kanban/chat over WebSockets, a Gemini 2.0 Flash assistant, JWT + Google OAuth RBAC, Dockerized CI/CD.

Flutter Developer Intern — Vatech WABAG

Summer 2023

Tunis, Tunisia

- Built a cross-platform employee leave-management app (REST API, secure auth with RBAC, Firebase real-time push).

PROJECTS

EDITH — Local-First Personal AI Companion (macOS)

Python, Swift, Ollama, MLX, LoRA, SQLite

100% local persistent AI companion: an always-alive daemon wakes a 9B model only when cognition is required (cold 20.7s / warm 0.36–0.77s to first token, idle under 1 GB). 1,596-test suite; base-model swaps gated on measured probes (10/10 instruction, 14/14 held-out honesty); LoRA fine-tuning on Apple MLX (0.06% trainable params, held-out val loss 3.89 → 2.73); independent replication of Anthropic’s agentic-misalignment methodology — 24 runs with pre-registered scoring, every scorer flag checked against the full transcript.

SentinelAI

LangGraph, FastAPI, Celery, Redis, Qdrant

Production multi-agent platform scoring geopolitical & market risk for Gold/Oil/S&P 500/BTC/ETH — a 7-agent LangGraph pipeline with Monte Carlo + GARCH risk, sub-30s end-to-end, with rate-limit handling (429 back-off honoring Retry-After) on Groq.

ZK-Attest

Circom, snarkjs, Solidity, Hyperledger Besu, Next.js

Privacy-preserving credential verification proving facts without revealing data — Groth16 zero-knowledge proofs, W3C Verifiable Credentials, and an ERC-5192 soulbound trust chain (GDPR by design).

Agentic Valley

Claude Agent SDK, TypeScript, VSCode API

VSCode extension visualizing multi-agent orchestration as a living pixel-art office — a main agent plus 6 specialized agents, each its own Claude Agent SDK session, with every Task-tool delegation animated live.

EDUCATION

National Engineering Degree, Computer Engineering (highest distinction) — ESPRIT, Tunis

2021 – 2026

Baccalaureate in Computer Science

2021

CERTIFICATIONS & LANGUAGES

Certifications: NVIDIA Generative AI with Diffusion Models • NVIDIA Applications of AI for Predictive Maintenance • Hashgraph Developer Course • AWS Academy Cloud Foundations (2025)

Languages: Arabic (Native) • French (Native) • English (C1)